

Who Captures the Value Generated in Cyprus?

A Recipient-Country Accounting of Domestic Value Retention and Foreign Leakage,
2010–2023

Diomides Mavroyiannis*

Data release v0.1, September 2026

Abstract

In official statistics Cyprus pays abroad primary income—wages, profits, interest and dividends—worth between 67% and 172% of its GDP each year. Most of it is income that only passes through Cypriot holding companies. We ask a narrower question: of the value produced in Cyprus, how much accrues to residents, how much to the rest of the world, and to whom? We split every euro of GDP by recipient and industry, combining CYSTAT’s supply–use and input–output tables, statistics on foreign-controlled firms (FATS), the balance of payments by paying sector, and a hand-built ownership graph. In 2023 at most 92.9% of Cypriot GDP accrued to residents (for the central foreign equity share); 7.1% (EUR 2.30 bn) accrued abroad, 54% of it as profits of foreign-owned firms. Retention ranged from 92.9% to 95.8% over 2010–2023 and has fallen since 2020. A sector-by-sector bridge accounts for the difference from the official outflow. Separately, the import content of final demand for Cypriot output rose from 21.4% to 40.7% between 2010 and 2022.

Keywords: national accounts; value added; foreign direct investment; special purpose entities; input–output; ownership networks; Cyprus.

JEL: E01, F23, F62, C67, O52.

1 Introduction

When a Cypriot restaurant pays a food-delivery platform a commission of EUR 20, how much of that money ends up abroad, and why? A common answer is “all of it, because the platform is foreign-owned.” That is the wrong reason. Ownership moves only the owner’s profit abroad. The rest of the commission pays Cypriot staff, Cypriot taxes and suppliers. Some of those suppliers are Cypriot; others are abroad. Revenue is not value, and a foreign owner has a claim on profit, not on revenue.

This paper applies that logic to a whole economy. We ask: *of the value generated by production in Cyprus, how much accrues to residents, and how much to other countries—through foreign workers,*

*Milestone Institute, Budapest. dmavroyiannis8@gmail.com. Data, code and an interactive dashboard: <https://github.com/diomavro/country-value-retention>. All numbers in this paper are generated by the repository’s pipeline from official sources; every figure can be traced to its source table.

foreign owners, foreign lenders, and EU institutions? We also ask a companion question: *of each euro spent on Cypriot output, how much pays for imported inputs, and from where?*

Cyprus is a demanding first case. Its official accounts record primary income paid abroad of EUR 34.9 bn in 2023, which is 108% of GDP (EUR 32.4 bn). In 2018–2021 the figure exceeded 150% of GDP. Taken at face value, Cyprus would be giving away more than it produces. It is not. Most of these flows are dividends and interest that Cypriot-registered holding companies receive from abroad and pass on to foreign owners, with little Cypriot production in between. In 2022, financial corporations other than banks—the sector that contains Cyprus’s special purpose entities (SPEs)—paid EUR 22.3 bn of the investment income that left the country. GNI, which nets these flows against receipts, was EUR 29.0 bn in 2023. Neither GNI nor the gross outflow answers our question, because both mix value produced in Cyprus with value produced elsewhere and routed through it.

We therefore build the accounting from the production side. We start from GDP by industry and split each industry’s value added into compensation of employees, taxes on production, and operating surplus. Each outflow may draw only on its own kind of income. Profits of foreign-owned firms come out of operating surplus, never out of wages or taxes, and pay to non-resident workers only out of compensation. Profits of foreign-owned firms are measured from FATS, which reports the operating surplus of foreign-*controlled* firms in Cyprus by industry and by the country of their ultimate controller. From that surplus we deduct depreciation, interest and Cypriot corporate tax, and apply the share of equity held by non-residents, calibrated on company filings. The result, $V[i, j, k, t]$, allocates every euro of Cypriot GDP in year t and industry k to one recipient j . Every cell carries its source, method, status and confidence.

Findings. At most 92.9% of Cyprus’s 2023 GDP accrued to residents. Given the central assumptions (section 11), it is an upper bound because any outflow the statistics miss is counted as retained. The remaining 7.1% (EUR 2.30 bn) accrued abroad: EUR 1,246 m as profits of foreign-owned firms, EUR 454 m as interest paid by firms and households to other foreign lenders (banks’ interest paid abroad was smaller than their receipts), EUR 145 m as interest paid to foreign parent companies, EUR 197 m as interest on public debt, EUR 182 m as pay to non-resident workers, and EUR 74 m as customs duties passed to the EU. These are *accruals*: profits that foreign owners reinvest in Cyprus still accrue to them, as in the balance of payments. Retention ranged between 92.9% (2023) and 95.8% (2020). It fell from 95.8% in 2020 to 92.9% in 2023 as the surplus of foreign-controlled firms grew. The largest identified recipients in 2023 were Greece (EUR 275 m), Offshore financial centres (EUR 250 m), United States (EUR 131 m) and Canada (EUR 109 m). The countries are those of the ultimate controller of the paying firm. 46% of the outflow cannot be assigned to a country because the statistics are confidential or published only as world totals; we report it as such.

Imports are a separate channel and are not part of GDP. Between 2010 and 2022, imported intermediate inputs grew from EUR 4.8 bn to EUR 18.7 bn. IT services, banking, auxiliary financial services and publishing account for EUR 8.7 bn of the EUR 13.9 bn increase. The import content of final demand for Cypriot output, counting suppliers’ suppliers, rose from 21.4% to 40.7%.

What is new. None of the ingredients is new; the contribution is to join them. Value-added-in-trade statistics say where value is *produced*. Foreign-affiliate statistics say which firms are foreign-*controlled*. The balance of payments says what crosses the border, but in a hub economy it is dominated by pass-through. We combine them into one accounting with four properties. It never counts an imported input and the foreign value embodied in it twice. It never counts wages or taxes as foreign capital income, and it counts interest once. It removes SPE income. And it is reconciled line by line with the official outflow. Every step is automated and published with its code.

What this paper does not claim. The accounting describes where income accrues. It does not say whether foreign ownership raised or lowered Cypriot incomes: a foreign-owned hotel that sends EUR 1m of profit abroad may also employ workers who would otherwise earn less. Leakage is an accounting concept, not a cost.

Section 2 explains the concepts with a worked example, and section 3 places the work in the literature. Sections 4 and 5 describe the data and methods. Sections 6 to 10 present results. Sections 11, 13 and 14 cover sensitivity, limitations and discussion.

2 Conceptual framework

A handful of ideas carry the whole analysis. We state each in plain terms first, with a small example; the numbers in this section are made up and do not enter the results.

Value added, not revenue. A bakery that sells EUR 100 of bread and buys EUR 40 of flour and EUR 10 of electricity has added EUR 50 of value. Summing revenue across firms double counts the flour, once in the miller’s sales and again in the baker’s. Summing value added gives GDP. If the flour was imported, the EUR 40 paid for something produced abroad, but it is the foreign miller’s value added, not “leakage of the bakery”.

Three kinds of income. The bakery’s EUR 50 is *primary income*: income earned by taking part in production. It pays three groups: workers (compensation of employees), the government (taxes on production, net of subsidies), and owners and lenders (gross operating surplus). Say EUR 30, EUR 2 and EUR 18.

Residence, not nationality. “Foreign” means *not resident*, the rule of the national accounts and the balance of payments [European Commission et al., 2009, International Monetary Fund, 2009]. A Greek citizen who lives and works in Nicosia is a Cypriot resident, so their wage stays in Cyprus. A worker who lives in Greece and comes for a short contract is not. We never use citizenship or country of birth as a proxy for residence. A company is resident where it is incorporated.

Owners get profit, lenders get interest. Operating surplus must first replace worn-out equipment (depreciation) and pay lenders (interest). Cypriot corporate tax is due on what remains. Only the rest is profit, and an owner’s claim is its share of that profit. In the bakery, take the EUR 18 of surplus, subtract EUR 5 of depreciation and EUR 3 of interest to a Cypriot bank, and

apply 12.5% tax: EUR 8.75 is left. A foreign owner of 60% of the shares has a claim on EUR 5.25, and the other EUR 44.75 of the EUR 50 stays in Cyprus. Had the bakery borrowed from a German bank, the EUR 3 of interest would accrue to Germany. It is counted once, as interest. The same host-tax logic applies to workers: the part of a non-resident's pay that goes to Cypriot social insurance stays in Cyprus.

Accruing is not paying. If the foreign owner leaves its EUR 5.25 in the bakery to buy a new oven, the profit has still accrued to the owner. The balance of payments records this as *reinvested earnings*. Our figures measure claims on value, not cash crossing the border in the year.

Following ownership up the chain. A Cypriot subsidiary may be owned by a Dutch holding company owned by a US parent. Official statistics on foreign-controlled firms attribute a firm to its *ultimate controlling institutional unit*: the unit at the top of the chain of majority control [OECD, 2008]. We use that concept for recipient countries. Where the records run out, we say *unresolved* rather than invent a country.

The chain behind the chain. A restaurant imports olive oil directly and buys bread from a bakery that imports flour; the second import is indirect. The input–output table records who buys from whom. Its Leontief inverse $L = (I - A)^{-1}$ adds up suppliers, suppliers' suppliers, and so on [Leontief, 1936, Miller and Blair, 2009]. Every euro of final spending on domestic output (valued before taxes on the final sale, such as VAT) is exhausted by three parts:

$$\underbrace{(v/x)L}_{\text{domestic value added}} + \underbrace{\mathbf{1}'A_mL}_{\text{imported inputs}} + \underbrace{(t/x)L}_{\text{net taxes on products used as inputs}} = \mathbf{1}'. \quad (1)$$

Imports and value added are two slices of the same euro, so they cannot be double counted.

Two frames. *Frame A* asks who receives the value added generated in Cyprus in a year: GDP minus what accrues to non-residents is retained. *Frame B* asks where a euro spent on a Cypriot product ends up: its Cypriot value added is routed through *Frame A*'s recipient shares, and its imported inputs go to the countries that supplied them.

Why GNI is not the answer. GNI adds income residents earn abroad, which is value generated elsewhere. And in a hub economy, primary income paid abroad includes income that SPEs receive from abroad and pass on. Neither belongs in an account of value generated by Cypriot production.

Codes used in the paper. D1 compensation of employees; D12 employers' social contributions; D29–D39 other taxes on production net of subsidies; B2A3G gross operating surplus and mixed income (the income of the self-employed); CFC consumption of fixed capital (depreciation); NOS net operating surplus; S11 non-financial corporations; S12 financial corporations, of which S122 banks and S12M other financial corporations (where SPEs sit); S13 government; S1V firms and households; S2 the rest of the world.

3 Related Literature

This paper joins three literatures that measure the same object—the destination of value added generated on a territory—from different sides, and that have so far been kept apart.

The first is the input–output literature on value added in trade. Building on [Leontief \[1936\]](#) and the domestic/imported coefficient split in [Miller and Blair \[2009\]](#), [Hummels et al. \[2001\]](#) measure the import content of exports, [Johnson and Noguera \[2012\]](#) define value-added exports, and [Koopman et al. \[2014\]](#) and [Los et al. \[2016\]](#) decompose gross exports into value-added and double-counted terms. [Los et al. \[2015\]](#) show that the foreign value-added share of final manufactures has risen almost everywhere. Surveys by [Johnson \[2018\]](#) and [Antràs and Chor \[2022\]](#) make clear that these measures, and the databases that implement them [[Timmer et al., 2015](#), [Yamano et al., 2021](#), [Eurostat, 2019](#), [Ahmad et al., 2017](#)], attribute value added to the country where production takes place. We borrow their double-counting logic but ask a different question: not where value is *produced*, but who *receives* it.

The second literature slices value chains by factor income. [Timmer et al. \[2013\]](#) introduce “GVC income”, [Timmer et al. \[2014\]](#) show that the capital share of value-chain income has risen, and [Timmer et al. \[2019\]](#) document functional specialisation across countries. [Chen et al. \[2018\]](#) attribute a large and growing residual to intangible capital and note that its location is recorded where intellectual property is registered. This is exactly the case in a holding-company jurisdiction such as Cyprus, and it motivates the step these papers leave out: allocating capital income to the residence of its owner.

A closely related strand splits input–output tables by firm ownership. The OECD AMNE database [[Cadestin et al., 2018](#)] and [Miroudot and Ye \[2020\]](#) separate value added of foreign affiliates from that of domestic firms. Extended supply–use tables for the United States [[Fetzer et al., 2018](#)] and the Netherlands [[Chong et al., 2019](#)], codified in [United Nations \[2023\]](#), provide the compilation template we follow. [Lipsey \[2010\]](#) and [Casella et al. \[2023\]](#) warn that when intangibles and conduit FDI dominate, neither location nor immediate ownership identifies where production income accrues. The ownership-split tables stop at foreign-*controlled* value added. They do not ask how much of it leaves the country, as distributed or reinvested earnings, or whether it only passes through on its way to a third country.

The third literature works from the residence side. The SNA, BPM6 and the FDI benchmark definition [[European Commission et al., 2009](#), [International Monetary Fund, 2009](#), [OECD, 2008](#)] define the primary-income flows that bridge GDP and GNI. [Rassier \[2017\]](#) and [Guvenen et al. \[2022\]](#) show that profit shifting by multinationals distorts output, BoP and productivity by location. Ireland’s modified GNI* [[Economic Statistics Review Group, 2016](#), [FitzGerald, 2018](#)] is the leading official response, but it is a single adjusted aggregate. It does not decompose the leakage by sector or by recipient. Our requirement that every euro of GDP be allocated to exactly one recipient is the cross-border counterpart of the distributional national accounts of [Piketty et al. \[2018\]](#).

Measuring these flows correctly requires dealing with special-purpose entities. Investment-income flows are the counterpart of external positions [[Lane and Milesi-Ferretti, 2018](#)], and those positions are heavily distorted by conduits. [Damgaard and Elkjaer \[2017\]](#) and [Damgaard et al. \[2024\]](#) estimate that a large share of global FDI is “phantom” investment in shell entities, and [Casella \[2019\]](#)

proposes a probabilistic look-through to ultimate investors. [Blanchard and Acalin \[2016\]](#) and [Bolwijn et al. \[2018\]](#) show that hub FDI is largely pass-through. [Zucman \[2013\]](#) and [Coppola et al. \[2021\]](#) demonstrate that restating external positions from residence to nationality changes bilateral patterns substantially. Cyprus appears on the haven lists used to estimate shifted profits [[Hines and Rice, 1994](#), [Tørsløv et al., 2023](#)]. Related work quantifies profit shifting from FDI returns [[Janský and Palanský, 2019](#)] and traces its footprint in the balance of payments [[Hebous et al., 2021](#)]. Network studies of corporate control [[Vitali et al., 2011](#)] classify Cyprus as a sink offshore financial centre [[Garcia-Bernardo et al., 2017](#)]. Russian round-tripping through Cyprus is documented by [Ledyaeva et al. \[2015\]](#) and [Repousis et al. \[2019\]](#), and IMF surveillance repeatedly flags the SPE-driven volatility of Cyprus’s external accounts [[International Monetary Fund, 2023, 2024](#)]. We follow the IMF definition of SPEs [[Sanchez-Muñoz et al., 2022](#)], as applied by the Central Bank of Cyprus [[Central Bank of Cyprus, 2026](#)], and use Eurostat’s SPE split of FDI income together with a resident-sector rule. This removes pass-through flows explicitly, instead of counting them either as domestic retention or as foreign capture.

Finally, for a tourism-intensive economy, the older tourism-leakage literature [[Sinclair, 1998](#)] and recent work on platform intermediation [[Rochet and Tirole, 2003](#), [Hunold et al., 2020](#), [UNCTAD, 2019](#)] point to a further channel. A commission paid to a foreign-owned platform is often read as value captured abroad, but only the platform owner’s share of profit is; the rest pays local staff, suppliers and taxes. The accounting gives each part an explicit recipient.

Relative to these literatures, our contribution is integration rather than a new estimator. We join the residence, ownership and value-chain layers into a single recipient-country by sector tensor. The tensor adds up to GDP, is consistent with BPM6 primary income and FATS ownership, excludes double counting between imported-input leakage and foreign capital-income leakage, and is adjusted for SPEs. Every cell is traceable to a published source.

4 Data

All inputs are official statistics, downloaded by scripts and logged with URL, retrieval date and SHA-256 checksum in a catalogue of 96 datasets (appendix [A](#)). Processing stops if a raw file no longer matches its checksum. Table [1](#) lists the core sources.

Two features of the data shape the method. First, the input–output tables are complete. The symmetric tables contain no confidential cells and reconcile exactly to GDP and compensation in every year (appendix [B](#)). Second, the balance of payments is dominated by pass-through. In 2023, FDI income paid abroad was EUR 30.7 bn. Excluding SPEs, as Eurostat publishes them, it was EUR 6.2 bn (19.0% of GDP). Even that exceeds the operating surplus of all foreign-controlled firms in FATS (EUR 2.76 bn; EUR 2.51 bn with finance at the national-accounts level), because holding and trading companies with staff are not SPEs but still pass on foreign-earned income. We therefore measure foreign-owned profit from the production side (FATS) and use the payments side as a comparator, bridged line by line in section [7](#).

Table 1: Core data sources

Source	Content used	Coverage
CYSTAT supply–use and symmetric input–output tables (0640010E–0640055E)	domestic and imported intermediate use, 65 products; value-added components	2010–2022
CYSTAT sector accounts (0630030E) and Eurostat <code>nasa_10_nf_tr</code>	rest-of-world primary income (D1, D2, D4); corporate depreciation, interest, employers’ contributions	2010–2024
Eurostat <code>nama_10_a64</code>	GVA, compensation, depreciation, operating surplus by industry	2010–2024
Eurostat FATS (<code>fats_activ</code> ; <code>fats_gla_08</code>)	operating surplus of foreign-controlled firms by industry and ultimate controlling country; before 2021 derived as value added minus personnel costs	2010–2023
Eurostat BoP (<code>bop_c6_a</code> , <code>bop_rem6</code> , <code>bop_fdi6_inc</code>)	investment income by paying resident sector; compensation by partner; FDI income, total and SPE	2010–2024
Eurostat FIGARO (<code>naio_10_fcp_ii*</code>)	supplying country of Cyprus’s imported intermediates	2010–2022
Company filings (42 operating firms)	shareholders, parents, ultimate controllers	2024–2026

Notes: confidential cells (14%–23% of FATS cells, 27% of FDI income cells up to 2023) are kept as missing, never set to zero. Where a missing BoP cell leaves a flow unmeasured, the estimate is a lower bound on that flow and the omission is reported. Full catalogue in `data_catalogue.csv`.

5 Methodology

5.1 The recipient accounting (Frame A)

For each industry k and year t , the national accounts give $GVA_k = D1_k + D29X39_k + B2A3G_k$. An outflow to country j through mechanism m is written O_{jk}^m . Each mechanism may draw only on its own component, and never more than that component:

$$\sum_j O_{jk}^{\text{labour}} \leq D1_k, \quad \sum_{j,m \in \text{capital}} O_{jk}^m \leq \max(B2A3G_k, 0), \quad (2)$$

and the pipeline stops if either bound is violated. Retained value is the residual, $V[\text{CY}, \text{CY}, k] = GVA_k - \sum_{j,m} O_{jk}^m$, so $\sum_{j,k} V = \text{GDP}$ by construction. This is an identity, not evidence of accuracy. Taxes on products form a pseudo-industry.

Profits of foreign-owned firms. FATS reports gross operating surplus of foreign-controlled firms, before depreciation, interest and tax. We use the finest published FATS code for each industry: for example publishing (J58) or IT services (J62–J63), and before 2021 every 2-digit division. Only a section’s unpublished remainder is split across its industries in proportion to operating surplus. We then deduct depreciation (c) and interest paid (ρ), both per euro of gross operating surplus of non-financial corporations. We use corporate ratios, not industry ratios, because industry surplus also includes the income of the self-employed. For ρ we use the interest non-financial corporations pay, after the national-accounts adjustment for bank services (FISIM), per euro of their surplus. We

do not net interest received: receipts from abroad are not value generated in Cyprus, and netting would credit them to foreign owners. Actual interest before the FISIM adjustment is much higher in the post-crisis years, because domestic firms were heavily indebted: applied to all foreign-controlled firms it would leave them no or almost no profit in 2010–2015 and 11–29% of surplus in 2016–2020 (in 2021–2023 it is close to, and in 2023 below, the central ratio). We report it, the net-of-receipts version and $\rho = 0$ as variants (section 11); the year-by-year ratios are in appendix B. We apply the statutory corporate tax rate τ_t and multiply by θ , the non-resident share of equity:

$$O_{jk}^{\text{FDI}} = \text{GOS}_k^{\text{FATS}} (1 - c - \rho) (1 - \tau_t) \theta \sigma_{s(k),j}. \quad (3)$$

On this basis ρ was 34% in 2010 and 27% in 2023. For finance, $\rho = 0$, because interest is the core business. τ_t is 10% to 2012 and 12.5% from 2013. Effective rates are lower, so the no-tax case is reported as a bound (section 11). The recipient share $\sigma_{s,j}$ is fitted section by section. From 2021, when FATS covers finance, its operating surplus of all financial firms can exceed what the national accounts record for finance; in financial centres business statistics can include income of holding and fund structures. Where it does, we keep the FATS foreign-controlled share of finance, and of each country within it, and take the level from the national accounts. We apply this to finance only: in other sections FATS and the national accounts differ by definition (before 2021 FATS surplus is value added minus personnel costs), for Cyprus in both directions, and a one-sided cut would bias foreign profit down. FATS publishes some section $\times$ country cells; these are held fixed, and the unpublished cells are fitted so that rows add to section totals and columns to the published country totals (iterative proportional fitting). Countries that Eurostat suppresses remain in a “confidential” category: we do not model values for suppressed cells, which could otherwise be backed out from published totals. Before 2021 FATS excludes finance, so profits of foreign-owned banks enter through the balance of payments. These are FDI equity income paid by deposit-taking corporations, which is already after interest and tax. Foreign-owned insurers and auxiliary financial firms are not covered before 2021 (section 13).

The foreign equity share θ . θ is not published. We calibrate it from the ownership graph as the integrated non-resident share of each documented foreign-controlled firm. Ownership is traced through Cypriot holding companies to the first non-resident owner on every path, so foreign minority holders count too. The 14 firms belong to 10 groups, identified by their first non-resident owner. Counting each group once gives $\theta = 0.90$. Firms are in the sample only if more than half of their equity is held by non-residents, so θ exceeds 0.5 by construction. Most are wholly owned subsidiaries, so θ close to 1 is essentially an assumption that foreign-controlled affiliates are wholly owned. We report results for $\theta \in [0.6, 1]$.

Interest, workers, government, EU. The interest deducted from foreign-controlled firms’ surplus goes to three kinds of lender. The part owed to Cypriot lenders stays in Cyprus. The part owed to foreign third-party lenders is already in the balance-of-payments interest paid by firms and households (S1V), so it is counted once, there. The part owed to the firms’ own foreign parents (FDI debt interest) is booked separately: we apply to the deducted interest the economy-wide share

of firms' interest that goes to foreign affiliates (s , 17% in 2010 and 33% in 2023), and allocate the result to industries in proportion to foreign-controlled surplus. Other interest paid abroad by firms and households on loans is split between firms and households in proportion to their interest paid; the household part, mostly mortgage interest but also business loans of the self-employed, is booked to imputed rent. Banks intermediate: they pay interest to non-resident depositors and earn it on foreign assets. We count their interest net of receipts and show the gross alternative, charged to borrowers rather than to the banks' own surplus. Pay to non-resident workers is the rest-of-world D1 net of employers' social contributions, which go to Cypriot social insurance (20.7% of compensation in 2023). Government interest to foreign creditors is attributed to the tax pool. Customs duties passed to the EU come from the rest-of-world D2 account. Income paid by financial corporations other than banks (S12M) is excluded as pass-through. The exception is the FATS profit of foreign-owned insurers and auxiliary financial firms. S12M is where the Central Bank of Cyprus classifies SPEs. Its operational criterion is at most three employees, little physical presence or production, control by non-residents, and transactions almost entirely with non-residents. The balance of payments by resident sector is published only for the world as a whole. These interest flows therefore have no partner country.

5.2 The input–output model (Frame B)

We use CYSTAT's symmetric product-by-product tables at basic prices, with the domestic block Z_d and the imported block Z_m kept separate. With $A_d = Z_d \text{diag}(x)^{-1}$, $A_m = Z_m \text{diag}(x)^{-1}$ and $L = (I - A_d)^{-1}$, equation (1) splits a final-demand vector f into three parts:

- domestic value added by industry, $\text{diag}(v/x)Lf$;
- imported inputs by product, $A_m Lf$;
- net taxes on products used as inputs.

Domestic value added in industry k is routed to recipients with Frame A's shares $V[\text{CY}, j, k]/\text{GVA}_k$. Imported inputs of product p go to supplying countries in proportion to FIGARO's supplies of p to Cypriot industries. That is the direct supplier, not necessarily the country where the value was added. We report two measures of foreign input exposure:

- *direct*: the imported share of intermediate inputs;
- *total*: $\mathbf{1}' A_m L$, the imports embodied in a euro of final demand through the whole chain.

They have different denominators and are not comparable by subtraction.

5.3 The ownership network

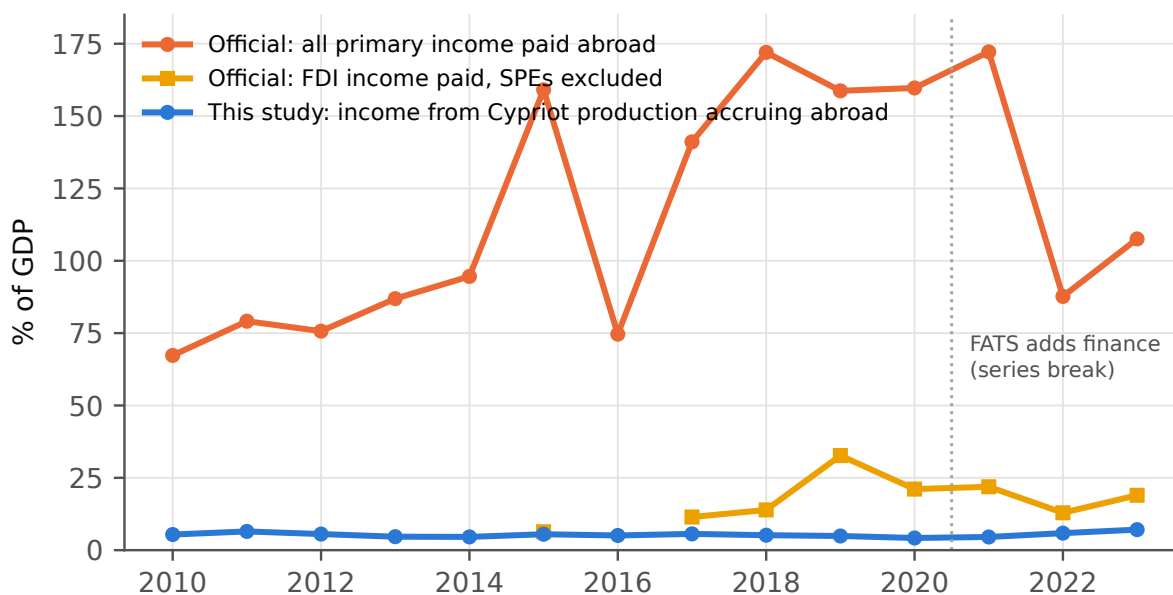
Entities and ownership links form a directed graph, and each link records its share, date, document URL, source tier and confidence. *Look-through* ownership solves $W = S(I - S)^{-1}$ for the direct-share matrix S , which assigns every unit of equity once, cross-holdings included; undocumented shares go to an explicit *unresolved* owner. *Control* follows majority voting rights, never dispersed free float,

up to a unit that nobody controls. A firm whose owners are undocumented, or whose majority is held by persons of undocumented residence, is *unresolved*, not presumed domestic. The company layer calibrates θ and illustrates ownership chains. Headline figures use official aggregates, because coverage of firms is partial.

5.4 Checks

Two kinds of automated checks run after every build (appendix B). *Identities* hold by construction, such as the accounting summing to GDP. *Source checks* recompute each outflow from its own raw source and fail if a flow is dropped, doubled or mis-scaled. Pay to non-resident workers must equal the rest-of-world D1 net of employers' contributions. The FATS base of foreign-owned profit must equal the published FATS total, less any amount above the national-accounts level. Foreign-owned banks must be counted exactly once. The bridge of section 7 must close. Unit tests pin the rules that revenue never enters, and that wages and taxes can never be booked as foreign capital income. They also forbid counting any flow or ownership link twice, and forbid listing aggregate partner codes such as the euro area beside their members.

6 Results: how much does Cyprus retain?

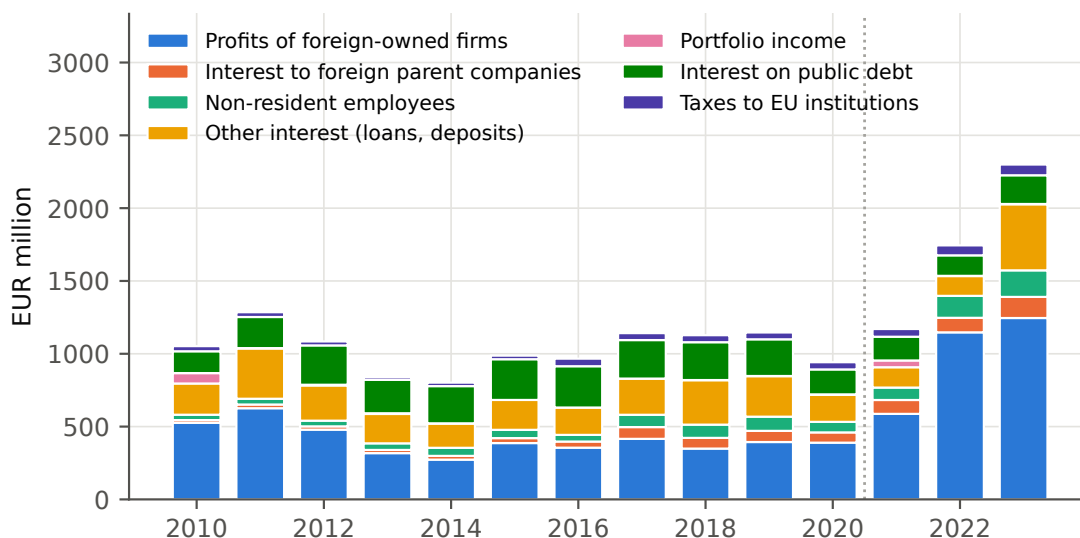


Sources: CYPSTAT sector accounts (S2); Eurostat bop_fdi6_inc, fats_activ, fats_g1a_08; author's calculations.

Figure 1: Official primary income paid abroad versus income from Cypriot production accruing abroad, % of GDP. The SPE-excluded FDI series starts in 2013 and has gaps where Eurostat marks cells confidential (2013–2014, 2016).

Figure 1 shows the central result. The official outflow—every euro of primary income paid to

non-residents, including pass-through—moves between 67% and 172% of GDP. FDI income paid excluding SPEs is far smaller but still volatile. Income generated by Cypriot production that accrues abroad lies between 4.2% and 7.1% of GDP, and has risen since 2020.



Foreign value leakage by mechanism (central estimate, $\theta = 0.90$, statutory corporate tax). Dotted line: FATS scope break.

Figure 2: Value accruing abroad by mechanism, EUR million.

Figure 2 decomposes the outflow. Profits of foreign-owned firms are the largest mechanism (54% in 2023), and they drive the increase after 2020. Pay to non-resident workers is small: 1.3% of compensation after employers' contributions. Interest paid abroad on public debt peaked at EUR 284 m in 2016, fell to EUR 141 m in 2022 and rose to EUR 197 m in 2023. The 2021 step coincides with FATS adding finance and sections P–R. With the same sections (B–N) in every year and banks from the balance of payments, retention in 2023 is 91.6%.

Two ratios summarise capital, both relative to the net operating surplus of all non-financial corporations, before interest and tax; the foreign-controlled numerator covers the same FATS sections (B–N) in every year. Foreign-controlled firms, weighted by their non-resident equity share, account for 22.0% of it in 2023 (8.5% in 2010). Because it is measured before interest, this share includes what those firms owe their lenders. Part of its rise coincides with the 2021 switch of the FATS dataset, which roughly doubled measured foreign-controlled surplus in section J. Interest and portfolio income that firms and banks paid to foreign lenders was equal in size to 7.6% of the same surplus; the two ratios overlap and are not added. These ratios have different bases from retention and are not added to it.

7 From the official outflow to this estimate

Table 2 bridges the official outflow to our estimate by paying sector. The largest line is income paid by non-bank financial corporations: EUR 29.2 bn in 2023, almost all of it pass-through. The

Table 2: From the official outflow to this study’s estimate, 2023 (EUR m)

Line (resident sector paying)	Official	This study	Reason for the difference
1 Compensation of employees paid (S2 D1)	230	182	model is net of Cypriot employer social contributions
2 Taxes on production to EU institutions (S2 D2)	74	74	identical
3 Property income: NA (S2 D4) minus BoP total (S1)	0	0	vintage/compilation gap between national accounts and BoP
4 Non-bank financial corporations incl. SPEs (S12M)	29,224	136	excluded as pass-through, except the FATS profit of foreign-owned insurers and auxiliary financial firms (from 2021)
5 Central bank (S121)	0	0	excluded: reserve management
6 Banks (S122)	1,500	500	model: foreign-owned finance profit (FATS K from 2021; BoP bank FDI equity income before) + banks’ interest net of receipts
7 General government (S13)	197	197	identical (interest items)
8a Firms and households: FDI equity income (S1V)	2,633	610	model uses FATS (production-based): the difference is income passed on by non-SPE holding and trading companies, plus FATS/BPM6 concept differences
8b Firms and households: FDI debt interest (S1V)	572	145	model caps intra-group interest at the interest deducted from foreign-controlled firms’ surplus
8c Firms and households: other interest and portfolio income (S1V)	457	456	identical where published; confidential items omitted (lower bound)
Total	34,887	2,299	official primary income paid vs this study’s outflow

Notes: official = CYSTAT rest-of-world account (D1, D2, D4 received by the rest of the world) and the balance of payments by paying resident sector (Eurostat bop_c6_a). The official lines add up to the official total in every year 2010–2023; the full series is in `data/processed/bridge.parquet`.

second is FDI equity income paid by firms and households. The balance of payments records EUR 2.63 bn, while the FATS-based profit of foreign-controlled non-financial firms is EUR 0.61 bn. The EUR 2.02 bn difference is income that holding and trading companies with staff receive from abroad and pass on, plus differences between the FATS and balance-of-payments concepts. Our data cannot split these two components. This line, not the SPE rule, explains most of the distance between the official figure and ours outside S12M.

Banks show the largest disagreement between two official sources. In 2023 the balance of payments records FDI equity income paid by banks of EUR 1,198 m, almost all of it reinvested earnings. Our FATS-based profit of foreign-owned banks is EUR 500 m, although the balance-of-payments figure is already after interest and tax and should, if anything, be smaller. The two sources agree in 2021 and 2022 (within EUR 52 m) and diverge only in 2023, when bank profits jumped. Our EUR 500 m is the banks’ share of FATS finance, which is published only as a whole; against all of finance the observed gap is smaller but still large. Three explanations are possible, and published data cannot separate them; the timing fits best a large bank that the two sources classify differently. FATS may treat Bank of Cyprus as domestically controlled, while the balance of payments treats all its profit as income of its Irish holding company. The balance of payments also includes 10–50% stakes, which FATS does not. And the two sources may record provisions differently. The ECB’s consolidated banking data are consistent with the first explanation in 2023: all banks in Cyprus earned EUR 1,272 m after tax, EUR 942 m in domestic banking groups and EUR 330 m in foreign-controlled subsidiaries and branches, and the balance-of-payments figure is close to the profit of all banks. This is weak evidence: in every year from 2013 to 2022 except 2018 the balance-of-payments figure exceeds the profit of all banks, consolidated group profit and FDI income differ in scope, and our FATS-based figure is above the ECB’s foreign-controlled profit, so the data do not settle the question. The central estimate uses FATS, the same source as every other industry, from 2021, and the balance of payments before 2021, when FATS does not cover finance: the series splices two sources that disagree in 2023. The consistent-scope series (91.6% retention in 2023) uses the balance of payments throughout, and shows the same post-2020 decline.

8 Industry decomposition

Table 3 reports, for industries of policy interest and the largest ones, their GVA, foreign input exposure, foreign ownership capture and retention. In 2023 the largest outflows came from Banking (EUR 512 m; retention 75.3%) and Publishing (incl. software) (EUR 227 m; 79.8%). Public administration retains almost all of its value added: it has little operating surplus to pay abroad. FATS publishes finance only as a whole, so foreign-owned profit in finance is split between banking, insurance and auxiliary services in proportion to their operating surplus; the split is an allocation, not an observation. Publishing (including software) and IT services are activities where intellectual property matters. Their measured surplus may partly be profit shifted into Cyprus by multinational groups [Guvenen et al., 2022]. Conversely, intra-group service imports can move profit out as intermediate consumption, which Frame A cannot see. For these industries “value generated by production” is an accounting label, not a statement about where the underlying activity takes place.

Table 3: Industry decomposition: GVA and retention 2023; input exposure 2022

Industry	GVA 2023 EUR m	Input exposure 2022 direct	total	Foreign own. 2023	Domestic retention 2023
Restaurants and accommodation	1,829	25.0	25.1	1.8	97.1
Retail	1,383	31.6	18.8	—	93.2
Banking	2,071	56.2	48.2	—	75.3
Insurance	162	51.5	44.8	—	84.6
Telecommunications	428	41.0	30.9	—	88.0
Energy	349	49.7	48.6	—	96.9
Land transport	200	45.5	37.1	—	95.9
Water transport (shipping)	480	47.0	46.5	—	93.2
Construction	1,595	20.1	37.0	—	98.0
Real estate (market)	1,142	23.9	7.8	—	95.9
Professional services	1,465	29.0	14.7	—	95.4
Manufacturing: food	435	26.8	40.3	—	96.2
Technology (IT services)	1,389	76.4	69.5	46.6	87.4
Publishing incl. software	1,124	71.8	62.2	37.5	79.8
Auxiliary financial services	703	81.3	67.8	—	83.2
Wholesale trade	1,686	41.8	15.8	—	91.1

Notes: percentages. Direct foreign input exposure = imported share of the industry's intermediate inputs. Total = imports embodied in one euro of final demand for the product, through the whole supply chain (Leontief inverse); the two have different denominators and are not comparable by subtraction. Foreign ownership capture = foreign-owned net operating surplus before interest and corporate tax (FATS, after corporate depreciation, times theta) as a share of the industry's net operating surplus, which also includes self-employed income. Domestic retention = share of the industry's GVA accruing to residents. From 2021 FATS publishes finance (K), trade (G) and most of transport (H) only at section level, so industries within these sections share one foreign share. Recipient countries are not shown by industry: FATS publishes the ultimate controlling economy mostly for whole sections, so industry-by-country cells are fitted, not observed (see the dashboard, flagged low confidence).

9 Country decomposition

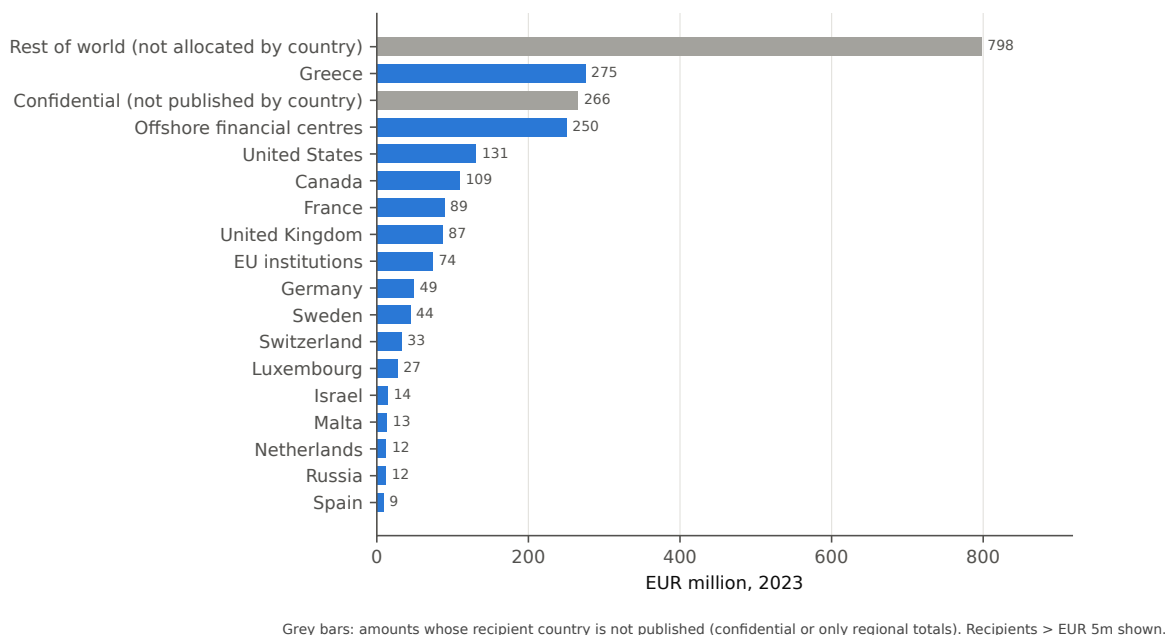
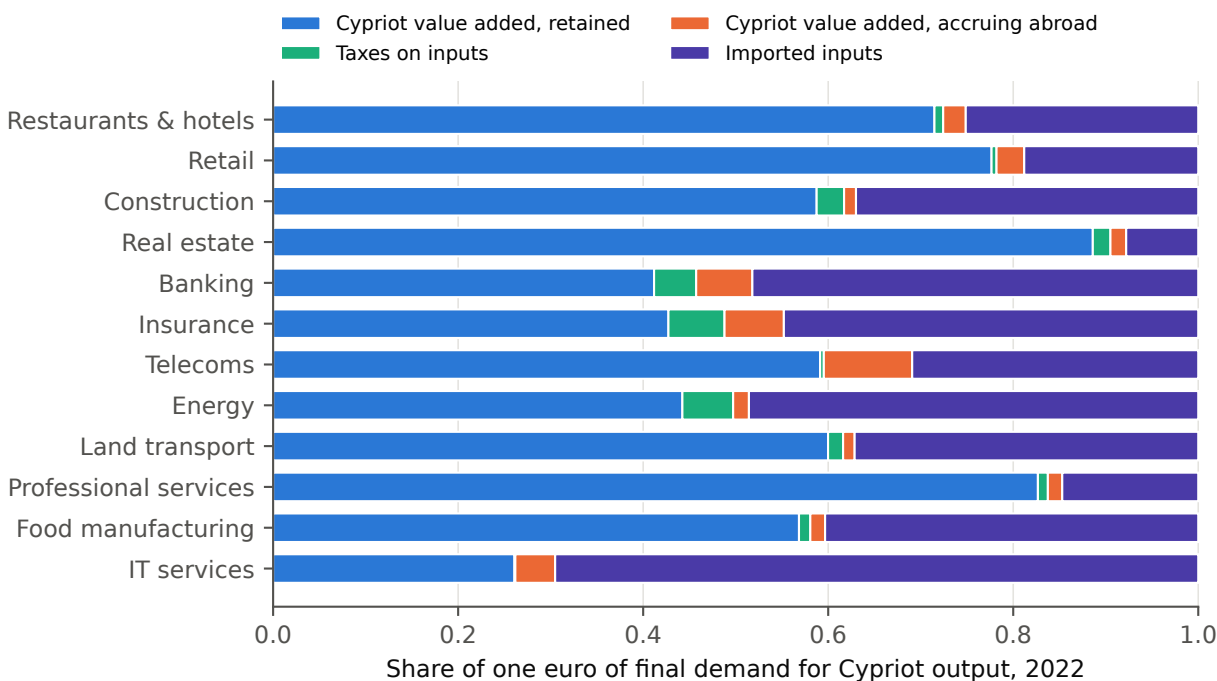


Figure 3: Recipients of value generated in Cyprus, 2023, EUR million, by ultimate controlling economy of the paying firm. Grey: country not published.

Figure 3 answers the question “to whom?”. The country is that of the ultimate controlling unit of the paying firm. For a listed parent, that is the parent’s country, even though its shareholders may live anywhere. Among identified recipients, Greece (EUR 275 m) and Offshore financial centres (EUR 250 m) lead, followed by United States, Canada and France. “Offshore financial centres” is Eurostat’s aggregate of offshore financial jurisdictions; it also contains operating economies such as Hong Kong, which we count only once, inside the aggregate. When FATS names them as ultimate controller, the chain stops there; statistics cannot see who stands behind an offshore vehicle. This is one reason why Russia, often discussed as a large investor in Cyprus, appears small as an ultimate controller of *producing* firms. Ireland does not appear as a named ultimate controller although the holding company of Bank of Cyprus is incorporated there. FATS flags some Irish cells in 2021–2023, finance included, as confidential, while it publishes empty cells as zero, so Irish-controlled firms exist in Cypriot finance. The published EU-controlled total less the published EU member cells shows that all confidential EU controllers of finance, Ireland included, together hold EUR 40 m to EUR 92 m a year, net; this caps Irish-controlled finance only if none of the other confidential cells is negative. We never assign a suppressed economy’s total to a named country. Within a country’s published total, however, the split across sections is fitted where the section cell is suppressed; those rows are flagged as fitted and carry low confidence. EUR 1,063 m (46%) cannot be assigned. Of the whole outflow, 26% goes to other EU countries and EU institutions and 28% outside the EU. EUR 266 m is suppressed for confidentiality, and EUR 798 m is interest paid by firms, households and the government that the balance of payments publishes only as a world total.

10 Value chains: where does a euro of spending go?



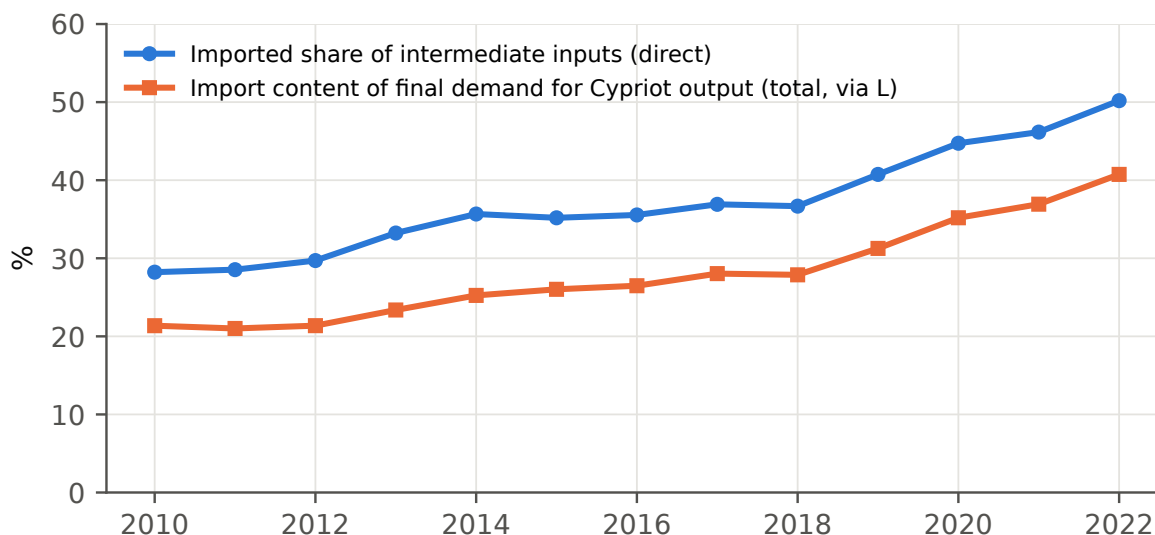
Leontief decomposition of CYPSTAT symmetric input-output table; imported inputs by direct supplier (FIGARO).

Figure 4: Where one euro of final demand for Cypriot output ends up, 2022 (basic prices).

Frame B traces spending (fig. 4). Of a euro spent in restaurants and hotels, 71 cents are Cypriot value added that stays in Cyprus, 2 cents are Cypriot value added accruing to foreign owners, workers or lenders, and 25 cents buy imported inputs. Retail keeps 78 cents, construction 59, and market real estate 89. IT services keep 26 cents: 70 cents of each euro buy imported inputs. Across all final demand for Cypriot output in 2022, Cypriot value added was EUR 26,301 m (equal to GVA, an identity) and imported inputs EUR 18,677 m. Among supplying countries, Russia (EUR 2.1 bn), Ireland (EUR 1.2 bn) and United States (EUR 1.2 bn) are the largest identified. FIGARO's residual "rest of the world" block is larger still, and its bilateral services trade is partly modelled. Russia's position as a leading supplier in 2022 reflects these modelled services flows and should be read with care.

Figure 5 shows that dependence on imported inputs rose steadily. The largest increases are in IT services, banking, auxiliary financial services and publishing.

A worked example (illustrative). Return to the EUR 20 commission. No accounts of the Cypriot subsidiary of the largest delivery platform are public. We give the platform the average cost structure of Cyprus's information-services industry and the ownership chain in company filings: the Cypriot company is owned by a Finnish company, which is owned by a US-listed parent. Most of the EUR 20 leaves Cyprus, but mainly as *imports*, not as the owner's profit. EUR 13.90 buys



Source: CYSTAT symmetric input-output tables 2010-2022 (0640050E, 0640055E); author's calculations.

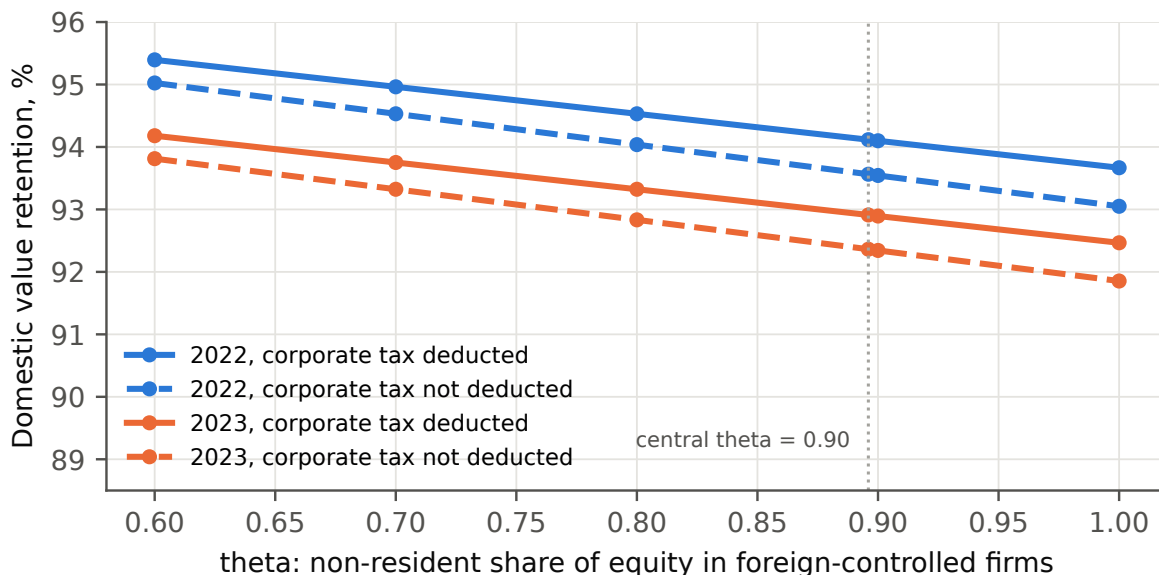
Figure 5: Foreign input exposure, direct and total, 2010–2022.

imported inputs such as software and cloud services, EUR 1.96 is profit accruing to the US parent, and EUR 0.16 is foreign income earned through Cypriot suppliers. EUR 3.98 stays in Cyprus, including EUR 2.03 of platform staff pay. The import share comes from the industry average, which is dominated by large IT firms buying services from affiliates abroad; a delivery platform may differ. The example is illustrative and is not part of any headline number.

11 Robustness and sensitivity

Figure 6 varies the two unobserved parameters. Moving θ from 0.6 to 1 changes 2023 retention by 1.7 percentage points. Not deducting corporate tax gives 92.4%. Using, from 2017, the European Commission's forward-looking effective tax rate on a new investment instead of the statutory rate gives 92.8%; earlier values appear to include taxes on immovable property (abolished in 2017), which the accounts already deduct. Using the FATS level of foreign-controlled finance surplus where it exceeds the national accounts gives 92.4%. Not deducting interest from foreign-controlled surplus gives 92.3%; deducting interest net of receipts gives 92.7%; deducting actual interest before the bank-services adjustment gives 92.8%. Three variants test whether retention is overstated:

- counting banks' interest paid abroad gross, not net of receipts, and charging it to the borrowers whose loans it funds, gives 92.0% in 2023 and 86.9% in 2010 (against 94.6%). Banks paid EUR 1,503 m of such interest in 2010 and received EUR 2,065 m;
- taking foreign-owned profit from the balance of payments instead of FATS, an upper bound because that series includes pass-through by holding companies, gives 86.9%;



Frame A re-estimated over the parameter grid; other assumptions at central values.

Figure 6: Domestic value retention under alternative θ and tax assumptions.

- assigning the outflows of non-bank financial corporations to Cypriot production, capped at that industry's surplus, gives 92.5%.

Across all variants, 2023 retention lies between 84.4% and 94.2%. In every variant, retention in 2023 is below its 2020 level; the year-to-year path in between differs across variants. Reconciliation is exact for GDP and compensation. The official data satisfy the GNI identity to within 0.008%, which checks the data we use, not the model.

12 Cyprus among other small open economies

Is 92.9% a lot or a little? To answer, we run the same estimator on five other euro-area economies: three that also host many foreign firms (Ireland, Luxembourg, the Netherlands) and two that do not (Greece, Portugal). Nothing in the method changes, only the sources. The rest-of-world account comes from Eurostat instead of CYSTAT, and the tax rate from the OECD. Run on Cyprus with the Eurostat sources, the estimator reproduces every mechanism to within the rounding of the two publications. Three caveats apply. First, θ is calibrated only for Cyprus, so elsewhere it is an assumption, and we also show $\theta = 1$; the two differ by at most 2.6 points. Second, where a country suppresses an input we coarsen or skip, and never fill it in. Malta could not be included because it never publishes mining and energy separately. Third, from 2021 FATS covers finance in every country, so in the hubs years before and after 2021 are not comparable; we compare countries within a year.

Table 4 shows three different ways to leak value. In Ireland, 21.9% of GDP in 2023 is profit of foreign-controlled firms, mostly in manufacturing and information and communication, and retention is 71.7%. In Luxembourg, the largest line, 17.5% of GDP, is pay to employees who live in France,

Table 4: Domestic value retention in Cyprus and five EU economies (% of GDP)

Country	Retention, Cyprus θ			$\theta = 1$	Official	Largest outflow
	2015	2019	2023	2023	outflow 2023	line, 2023
Cyprus	94.5	95.1	92.9	92.5	107.5	foreign-owned profit (3.8)
Ireland	79.8	78.6	71.7	69.2	73.9	foreign-owned profit (21.9)
Luxembourg	80.6	79.0	74.0	73.1	442.8	non-resident employees (17.5)
Netherlands	92.3	–	92.8	92.5	38.7	foreign-owned profit (2.7)
Greece	96.3	96.2	95.4	95.2	7.1	public debt interest (2.2)
Portugal	94.6	95.5	95.4	95.1	8.2	foreign-owned profit (2.8)

Notes: Frame A with the Cyprus method; other countries from Eurostat (`nasa_10_nf_tr`, `nama_10_a64`, FATS, `bop_c6_a`, `bop_rem6`) and OECD statutory tax rates. θ is calibrated for Cyprus only and assumed elsewhere; $\theta = 1$ gives the lowest retention. Official outflow: primary income paid abroad (S2), which includes pass-through income. – = inputs suppressed (Netherlands 2019). Malta is omitted in every year (sections B, D not published separately in the national accounts). From 2021 FATS covers finance; before, foreign-owned banks’ profit comes from the balance of payments, so columns before and after 2021 differ in scope. Suppressed balance-of-payments items are omitted (a suppressed bank receipt is not netted): Cyprus: portfolio dividends (firms and households) (2015, 2019, 2023), portfolio dividends (banks, received) (2015, 2019, 2023); Ireland: portfolio dividends (firms and households) (2015, 2019), portfolio dividends (banks, received) (2019, 2023); Luxembourg: foreign-owned banks’ profit (banks) (2015, 2019), loan and deposit interest (firms and households) (2015, 2019, 2023), portfolio dividends (firms and households) (2015, 2019, 2023), policyholder income (firms and households) (2015, 2019, 2023), intra-group interest (firms and households) (2015, 2019, 2023), loan and deposit interest (banks) (2015, 2019, 2023), portfolio dividends (banks) (2015, 2019, 2023), policyholder income (banks) (2015, 2019, 2023), loan and deposit interest (banks, received) (2015, 2019, 2023), portfolio dividends (banks, received) (2015, 2019, 2023), policyholder income (banks, received) (2015, 2019, 2023), loan and deposit interest (government) (2015, 2019, 2023); Greece: foreign-owned banks’ profit (banks) (2015, 2019).

Belgium and Germany. They are non-residents, so their pay leaves the economy. Their nationality plays no part in this; where they live is what counts. Luxembourg’s retention of 74.0% is overstated: much of the interest and dividends paid abroad by its firms, households, banks and government is confidential and therefore missing. In Greece (95.4%), the largest line is interest on public debt held abroad, 2.2% of GDP.

The official outflow column is the reason for this exercise. Primary income paid abroad is 443% of GDP in Luxembourg, 108% in Cyprus, 74% in Ireland and 39% in the Netherlands. Most of it is income passing through holding companies and funds, so the official ratio says little about value retained. Ireland has a lower official ratio than Cyprus but retains much less, and the Netherlands a lower ratio than Cyprus and retains about as much. Cyprus retains about as much as the Netherlands (92.8%), less than Portugal (95.4%) and Greece, and at least 18 points more than Ireland and Luxembourg.

13 Limitations

The full list, with the data that would remove each limitation, is in the repository’s limitations file. The most important are these. (i) Retention is an upper bound. Outflows we cannot measure count as retained: rent earned by non-resident owners of Cypriot dwellings, and FDI income from stakes of 10–50% in domestically controlled firms. For the first, recorded FDI income paid by real estate, the industry in which dwellings owned by non-residents are recorded, is about EUR 12 m a year or less

in absolute value from 2013 on (net of reverse investment); whether all such rent is recorded there is not documented, so this bounds only what is recorded. The second cannot be separated in published data: FDI income paid by non-SPE enterprises was EUR 6.2 bn in 2023, but it mixes minority stakes with income that holding companies with staff pass on; the balance-of-payments variant of section 11 bounds both. (ii) Before 2021, foreign-owned insurers and auxiliary financial firms are missing. In 2021, their first year, they accounted for about 0.3% of GDP (an allocated share of FATS finance). (iii) θ is calibrated on documented large firms, not measured. (iv) Excluding non-bank financial corporations is a sector rule, not an entity-level SPE flag. (v) Residence follows incorporation, so a Cypriot bank whose holding company is incorporated in Ireland is Irish-controlled at company level, although its shareholders include Cypriots. (vi) FATS operating surplus and national-accounts operating surplus differ in concept (treatment of financial intermediation services, software and research, statistical units), and the interest deduction is not applied to finance. (vii) Imports are attributed to the direct supplier.

14 Discussion

The headline—Cyprus retains at most 92.9% of the value it produces—is less dramatic than the official outflow and more informative. It separates two things that official aggregates mix. One is Cyprus as a financial conduit, through which foreign income flows of the order of GDP (between two-thirds of GDP and 1.7 times GDP a year). The other is Cyprus as a producing economy, where foreign-controlled firms account for 22.0% of non-financial corporate operating surplus before interest and tax, a share that has grown since 2020. Adding a country is a data run, not a new model: every Eurostat input exists for all EU member states, and the only Cyprus-specific loader, for sector accounts, has a Eurostat equivalent.

15 Conclusion

Of the value generated in Cyprus in 2023, at most 92.9% accrued to residents and at least 7.1% to the rest of the world. Most of what accrued abroad was profits of foreign-owned firms, with Greece, Offshore financial centres, United States and Canada as the largest identified recipients. Official primary-income outflows, at 108% of GDP, measure something else: the traffic of a financial hub. Every number above can be traced in the accompanying database and dashboard, from the headline through industry, mechanism and source table to the calculation that produced it.

References

Nadim Ahmad, Timon Bohn, Nanno Mulder, Marcel Vaillant, and Dayna Zaclicever. Indicators on global value chains: A guide for empirical work. OECD Statistics Working Papers 2017/08, OECD Publishing, Paris, 2017.

Pol Antràs and Davin Chor. Global value chains. In Gita Gopinath, Elhanan Helpman, and Kenneth

- Rogoff, editors, *Handbook of International Economics*, volume 5, pages 297–376. Elsevier, 2022. doi: 10.1016/bs.hesint.2022.02.005.
- Olivier Blanchard and Julien Acalin. What does measured FDI actually measure? Policy Brief 16-17, Peterson Institute for International Economics, Washington, DC, 2016. URL <https://www.piie.com/publications/policy-briefs/what-does-measured-fdi-actually-measure>.
- Richard Bolwijn, Bruno Casella, and Davide Rigo. An FDI-driven approach to measuring the scale and economic impact of BEPS. *Transnational Corporations*, 25(2):107–143, 2018. doi: 10.18356/c4f9fd3c-en.
- Charles Cadestin, Koen De Backer, Isabelle Desnoyers-James, Sébastien Miroudot, Davide Rigo, and Ming Ye. Multinational enterprises and global value chains: The OECD analytical AMNE database. OECD Trade Policy Papers 211, OECD Publishing, Paris, 2018.
- Bruno Casella. Looking through conduit FDI in search of ultimate investors – a probabilistic approach. *Transnational Corporations*, 26(1):109–149, 2019. doi: 10.18356/8a8b094c-en.
- Bruno Casella, Maria Borga, and Konstantin Wacker. Measuring multinational production with foreign direct investment statistics: Recent trends, challenges, and developments. IMF Working Paper 2023/113, International Monetary Fund, Washington, DC, 2023.
- Central Bank of Cyprus. External sector statistics: Balance of payments, international investment position and foreign direct investment. Online statistical release, 2026. URL <https://www.centralbank.cy/en/statistics/external-sector-statistics>. Accessed 24 September 2026.
- Wen Chen, Bart Los, and Marcel P. Timmer. Factor incomes in global value chains: The role of intangibles. NBER Working Paper 25242, National Bureau of Economic Research, 2018.
- Stephen Chong, Rutger Hoekstra, Oscar Lemmers, Ilke Van Beveren, Marcel Van Den Berg, Ron Van Der Wal, and Piet Verbiest. The role of small- and medium-sized enterprises in the Dutch economy: An analysis using an extended supply and use table. *Journal of Economic Structures*, 8:8, 2019. doi: 10.1186/s40008-019-0139-1.
- Antonio Coppola, Matteo Maggiori, Brent Neiman, and Jesse Schreger. Redrawing the map of global capital flows: The role of cross-border financing and tax havens. *The Quarterly Journal of Economics*, 136(3):1499–1556, 2021. doi: 10.1093/qje/qjab014.
- Jannick Damgaard and Thomas Elkjaer. The global FDI network: Searching for ultimate investors. IMF Working Paper 17/258, International Monetary Fund, Washington, DC, 2017.
- Jannick Damgaard, Thomas Elkjaer, and Niels Johannesen. What is real and what is not in the global FDI network? *Journal of International Money and Finance*, 140:102971, 2024. doi: 10.1016/j.jimonfin.2023.102971.
- Economic Statistics Review Group. Report of the economic statistics review group (ESRG). Technical report, Central Statistics Office, Dublin, 2016. URL <https://www.cso.ie/en/media/>

[csoie/newsevents/documents/reportoftheeconomicstatisticsreviewgroup/Economic_Statistics_Review_\(ESRG\)_Report_Dec_2016.pdf](https://csoie/newsevents/documents/reportoftheeconomicstatisticsreviewgroup/Economic_Statistics_Review_(ESRG)_Report_Dec_2016.pdf).

European Commission, International Monetary Fund, OECD, United Nations, and World Bank. *System of National Accounts 2008*. United Nations, New York, 2009. doi: 10.18356/4fa11624-en.

Eurostat. European Union inter-country supply, use and input-output tables – full international and global accounts for research in input-output analysis (FIGARO). Eurostat Statistical Working Papers KS-TC-19-002, Publications Office of the European Union, Luxembourg, 2019. URL <https://ec.europa.eu/eurostat/web/products-statistical-working-papers/-/ks-tc-19-002>.

James Fetzer, Tina Highfill, Kassu Hossiso, Thomas Howells, III, Erich Strassner, and Jeffrey Young. Accounting for firm heterogeneity within U.S. industries: Extended supply-use tables and trade in value added using enterprise and establishment level data. NBER Working Paper 25249, National Bureau of Economic Research, 2018.

John FitzGerald. National accounts for a global economy: The case of Ireland. Quarterly economic commentary special article, Economic and Social Research Institute, Dublin, 2018.

Javier Garcia-Bernardo, Jan Fichtner, Frank W. Takes, and Eelke M. Heemskerk. Uncovering offshore financial centers: Conduits and sinks in the global corporate ownership network. *Scientific Reports*, 7:6246, 2017. doi: 10.1038/s41598-017-06322-9.

Fatih Guvenen, Raymond J. Mataloni, Jr., Dylan G. Rassier, and Kim J. Ruhl. Offshore profit shifting and aggregate measurement: Balance of payments, foreign investment, productivity, and the labor share. *American Economic Review*, 112(6):1848–1884, 2022. doi: 10.1257/aer.20190285.

Shafik Hebous, Alexander Klemm, and Yuou Wu. How does profit shifting affect the balance of payments? IMF Working Paper 2021/041, International Monetary Fund, Washington, DC, 2021.

James R. Hines, Jr. and Eric M. Rice. Fiscal paradise: Foreign tax havens and American business. *The Quarterly Journal of Economics*, 109(1):149–182, 1994. doi: 10.2307/2118431.

David Hummels, Jun Ishii, and Kei-Mu Yi. The nature and growth of vertical specialization in world trade. *Journal of International Economics*, 54(1):75–96, 2001. doi: 10.1016/S0022-1996(00)00093-3.

Matthias Hunold, Reinhold Kesler, and Ulrich Laitenberger. Rankings of online travel agents, channel pricing, and consumer protection. *Marketing Science*, 39(1):92–116, 2020. doi: 10.1287/mksc.2019.1167.

International Monetary Fund. *Balance of Payments and International Investment Position Manual*. International Monetary Fund, Washington, DC, 6th edition, 2009. doi: 10.5089/9781589068124.069.

International Monetary Fund. Cyprus: Selected issues. IMF Staff Country Report 23/193, International Monetary Fund, Washington, DC, 2023.

- International Monetary Fund. Cyprus: 2024 Article IV consultation – press release; staff report; and statement by the executive director for Cyprus. IMF Staff Country Report 24/137, International Monetary Fund, Washington, DC, 2024.
- Petr Janský and Miroslav Palanský. Estimating the scale of profit shifting and tax revenue losses related to foreign direct investment. *International Tax and Public Finance*, 26(5):1048–1103, 2019. doi: 10.1007/s10797-019-09547-8.
- Robert C. Johnson. Measuring global value chains. *Annual Review of Economics*, 10:207–236, 2018. doi: 10.1146/annurev-economics-080217-053600.
- Robert C. Johnson and Guillermo Noguera. Accounting for intermediates: Production sharing and trade in value added. *Journal of International Economics*, 86(2):224–236, 2012. doi: 10.1016/j.jinteco.2011.10.003.
- Robert Koopman, Zhi Wang, and Shang-Jin Wei. Tracing value-added and double counting in gross exports. *American Economic Review*, 104(2):459–494, 2014. doi: 10.1257/aer.104.2.459.
- Philip R. Lane and Gian Maria Milesi-Ferretti. The external wealth of nations revisited: International financial integration in the aftermath of the global financial crisis. *IMF Economic Review*, 66(1): 189–222, 2018. doi: 10.1057/s41308-017-0048-y.
- Svetlana Ledyeva, Päivi Karhunen, Riitta Kosonen, and John Whalley. Offshore foreign direct investment, capital round-tripping, and corruption: Empirical analysis of Russian regions. *Economic Geography*, 91(3):305–341, 2015. doi: 10.1111/ecge.12093.
- Wassily W. Leontief. Quantitative input and output relations in the economic systems of the United States. *The Review of Economics and Statistics*, 18(3):105–125, 1936. doi: 10.2307/1927837.
- Robert E. Lipsey. Measuring the location of production in a world of intangible productive assets, FDI, and intrafirm trade. *Review of Income and Wealth*, 56(s1), 2010. doi: 10.1111/j.1475-4991.2010.00385.x.
- Bart Los, Marcel P. Timmer, and Gaaitzen J. de Vries. How global are global value chains? A new approach to measure international fragmentation. *Journal of Regional Science*, 55(1):66–92, 2015. doi: 10.1111/jors.12121. Published online 2014.
- Bart Los, Marcel P. Timmer, and Gaaitzen J. de Vries. Tracing value-added and double counting in gross exports: Comment. *American Economic Review*, 106(7):1958–1966, 2016. doi: 10.1257/aer.20140883.
- Ronald E. Miller and Peter D. Blair. *Input-Output Analysis: Foundations and Extensions*. Cambridge University Press, Cambridge, 2nd edition, 2009. doi: 10.1017/CBO9780511626982.
- Sébastien Miroudot and Ming Ye. Multinational production in value-added terms. *Economic Systems Research*, 32(3):395–412, 2020. doi: 10.1080/09535314.2019.1701997. Published online 2019.

- OECD. *OECD Benchmark Definition of Foreign Direct Investment*. OECD Publishing, Paris, 4th edition, 2008. doi: 10.1787/9789264045743-en.
- Thomas Piketty, Emmanuel Saez, and Gabriel Zucman. Distributional national accounts: Methods and estimates for the United States. *The Quarterly Journal of Economics*, 133(2):553–609, 2018. doi: 10.1093/qje/qjx043.
- Dylan G. Rassier. Improving the SNA treatment of multinational enterprises. *Review of Income and Wealth*, 63(s2), 2017. doi: 10.1111/roiw.12323.
- Spyridon Repousis, Petros Lois, and Pavlos Kougioumtsidis. Foreign direct investments and round tripping between Cyprus and Russia. *Journal of Money Laundering Control*, 22(3):442–450, 2019. doi: 10.1108/JMLC-08-2018-0054.
- Jean-Charles Rochet and Jean Tirole. Platform competition in two-sided markets. *Journal of the European Economic Association*, 1(4):990–1029, 2003. doi: 10.1162/154247603322493212.
- Carlos Sanchez-Muñoz, Artak Harutyunyan, and Padma Hurree Gobin. Special purpose entities: Guidelines for a data template. IMF Technical Notes and Manuals 2022/006, International Monetary Fund, Washington, DC, 2022.
- M. Thea Sinclair. Tourism and economic development: A survey. *Journal of Development Studies*, 34(5):1–51, 1998. doi: 10.1080/00220389808422535.
- Marcel P. Timmer, Bart Los, Robert Stehrer, and Gaaitzen J. de Vries. Fragmentation, incomes and jobs: An analysis of European competitiveness. *Economic Policy*, 28(76):613–661, 2013. doi: 10.1111/1468-0327.12018.
- Marcel P. Timmer, Abdul Azeez Erumban, Bart Los, Robert Stehrer, and Gaaitzen J. de Vries. Slicing up global value chains. *Journal of Economic Perspectives*, 28(2):99–118, 2014. doi: 10.1257/jep.28.2.99.
- Marcel P. Timmer, Erik Dietzenbacher, Bart Los, Robert Stehrer, and Gaaitzen J. de Vries. An illustrated user guide to the world input–output database: The case of global automotive production. *Review of International Economics*, 23(3):575–605, 2015. doi: 10.1111/roie.12178.
- Marcel P. Timmer, Sébastien Miroudot, and Gaaitzen J. de Vries. Functional specialisation in trade. *Journal of Economic Geography*, 19(1):1–30, 2019. doi: 10.1093/jeg/lby056. Published online 2018.
- Thomas Tørsløv, Ludvig Wier, and Gabriel Zucman. The missing profits of nations. *The Review of Economic Studies*, 90(3):1499–1534, 2023. doi: 10.1093/restud/rdac049. Published online 2022.
- UNCTAD. *Digital Economy Report 2019: Value Creation and Capture – Implications for Developing Countries*. United Nations, Geneva, 2019. doi: 10.18356/c7dc937a-en.

United Nations. *Accounting for Global Value Chains: GVC Satellite Accounts and Integrated Business Statistics*. Studies in Methods, Series F. United Nations, New York, 2023. doi: 10.18356/9789210048965.

Stefania Vitali, James B. Glattfelder, and Stefano Battiston. The network of global corporate control. *PLoS ONE*, 6(10):e25995, 2011. doi: 10.1371/journal.pone.0025995.

Norihiko Yamano, Joaquim Guilhoto, Ali Alsamawi, Colin Webb, Peter Horvát, Agnès Cimper, Carmen Zürcher, and Xue Han. Development of the OECD inter country input-output database 2021. Oecd science, technology and industry working papers, OECD Publishing, Paris, 2021.

Gabriel Zucman. The missing wealth of nations: Are Europe and the U.S. net debtors or net creditors? *The Quarterly Journal of Economics*, 128(3):1321–1364, 2013. doi: 10.1093/qje/qjt012.

A Data appendix

Every dataset is listed in `data_catalogue.csv` with provider, URL, reference period, publication and download dates, licence, scope, resolution, variables, processing script and SHA-256 checksum. Raw files are not redistributed; `make ingest` downloads them again, and checksums confirm that the same vintage is used.

Table 5: Datasets in the catalogue by provider

Provider	Datasets
CYSTAT (Statistical Service of Cyprus)	43
Central Bank of Cyprus	8
Compiled from company filings (Tier 1-4, per row)	1
Eurostat	44

B Methodological appendix

The validation suite (`python -m cvr.validate`) runs after every model build and exits with an error if any check fails. In this release 30 of 30 checks pass; unit tests add further guards on synthetic data.

Table 6: Depreciation, interest and intra-group share per euro of non-financial corporations' surplus

Year	c	ρ (central)	ρ net	ρ pre-FISIM	s
2010	0.34	0.34	0.29	0.67	0.17
2011	0.34	0.38	0.33	0.66	0.17
2012	0.38	0.33	0.28	0.70	0.19
2013	0.38	0.35	0.27	0.72	0.17
2014	0.43	0.34	0.23	0.74	0.21
2015	0.41	0.45	0.33	0.76	0.17
2016	0.34	0.36	0.28	0.55	0.21
2017	0.32	0.30	0.25	0.48	0.40
2018	0.30	0.28	0.25	0.42	0.37
2019	0.30	0.28	0.21	0.41	0.41
2020	0.34	0.27	0.21	0.45	0.42
2021	0.29	0.19	0.13	0.31	0.49
2022	0.26	0.12	0.07	0.20	0.57
2023	0.26	0.27	0.19	0.21	0.33

Notes: ratios to S11 gross operating surplus (Eurostat `nasa_10_nf_tr`). Central ρ = interest paid after the FISIM adjustment; net = paid minus received; pre-FISIM = actual interest paid. s = FDI debt interest paid by firms and households (BoP) / S11 interest paid.

Table 7: Reconciliation with official aggregates

Check	Years	Max diff. %	Tol. %	Pass
Compensation of employees (SIOT vs sector accounts S1 D1 paid)	2010–2022	0.0001	0.50	✓
GDP = sum of GVA + taxes less subsidies on products	2010–2022	0.0002	0.50	✓
GNI = GDP + primary income received - primary income paid (rest of world)	2010–2024	0.0077	0.10	✓
Recipient tensor sums to GDP	2010–2023	0.0005	0.50	✓

Notes: official values from CYSTAT annual national accounts (April 2026 vintage) and sector accounts; model values from the input-output tables and the recipient accounting. Residuals are reported, not forced to zero.

Table 8: Automated validation checks

Check	Pass
[identity] tensor sums to GDP every year	✓
[identity] industry rows sum to GVA (all years)	✓
no negative outflows	✓
no negative unexplained retained flows	✓
[identity] $DVR + FVL = 1$	✓
$0 \leq FVL \leq 1$	✓
[identity] Frame B: each euro sums to 1	✓
[identity] Frame B: domestic VA of all final demand = GVA	✓
Frame B: imported inputs counted once (= total imported intermediates)	✓
IO: $VA + import + tax\ content = 1$ (every product, every year)	✓
IO: direct import share within $[0,1]$	✓
ownership: no duplicate edges, shares $\leq 100\%$	✓
ownership: ultimate shares sum to 100% per firm	✓
reconciliation: all checks within tolerance	✓
provenance complete on all tables	✓
intra-group interest only on industries with foreign-controlled surplus	✓
source: creditor share = firms' and banks' interest abroad / S11 NOS	✓
fitted-cell flag and low confidence exactly where the FATS cell is suppressed	✓
source: banks' interest = $\max(\text{BoP paid} - \text{received}, 0)$	✓
source: households' loan interest on imputed rent = their D41 share x S1V loan interest	✓
source: every FATS row = $\text{base} \times (1 - \text{CFC} - \text{interest}) \times (1 - \text{tau}) \times \text{theta}$ (raw S11/S12 ratios)	✓
theta used = theta recalibrated from the company layer	✓
source: intra-group interest = $\min(\text{BoP FDI debt interest}, s \times \text{interest deducted})$	✓
source: non-resident pay = $S2\ D1 \times (1 - \text{employer SSC share})$	✓
source: EU taxes = S2 D2	✓
source: public-debt interest = BoP S13 interest paid	✓
source: FATS base of foreign-owned profit = published FATS total (less losses)	✓
source: foreign-owned banks counted once (BoP before 2021, FATS K from 2021)	✓
bridge: official lines add up to official outflow; model lines to model outflow	✓
time-series breaks identified and documented	✓